

$$p(0) = \frac{1}{2} \int_a^b (x(t)y'(t) - x'(t)y(t)) dt$$

sklenjena krivulja:

$$p(0) = \int_a^b x(t)y'(t) dt = - \int_a^b x'(t)y(t) dt$$

(2.)

$$\vec{r}(t) = (a \cos^3 t, a \sin^3 t)$$



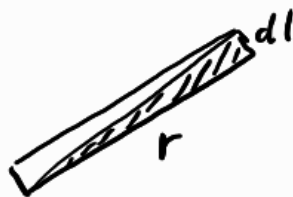
$$p(0) = \frac{3}{2} a^2 \int_0^{2\pi} \cos^2 t \sin^2 t dt = \frac{3}{2} a^2 \int_0^{2\pi} \frac{1}{4} \sin^2(2t) dt = \frac{3}{16} a^2 \int_0^{2\pi} (1 - \cos 4t) dt$$

$$= \frac{3}{16} a^2 \left(t - \frac{\sin 4t}{4} \right) \Big|_0^{2\pi} = \frac{3}{16} a^2 2\pi = \underline{\underline{\frac{3}{8} \pi a^2}}$$

polarno:

$$S(0) = \frac{1}{2} \int_{\alpha}^{\beta} \underbrace{r(\varphi)^2}_{= r(\varphi) \cdot r(\varphi)} d\varphi$$

$$= r(\varphi) \cdot r(\varphi) d\varphi = r(\varphi) \cdot dl$$



$$\frac{1}{2} r \cdot dl$$

③. $S = ?$

$$r(\varphi) = a(1 + \cos \varphi) \quad \varphi \in [0, 2\pi]$$

kardioida

$$S(D) = \int_0^\pi a^2(1 + 2\cos \varphi + \cos^2 \varphi) d\varphi = a^2 \left(\left(\varphi + 2\sin \varphi \right) \Big|_0^\pi + \int_0^\pi \frac{1 + \cos 2\varphi}{2} d\varphi \right)$$

$$= a^2 \left(\pi + \frac{\pi}{2} \right) = \underline{\underline{\frac{3\pi}{2} a^2}}$$

Dolžina krivulje:

$$\vec{r}(t) = (x(t), y(t))$$

$$L = \int_a^b \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2} dt$$

graf: $f[a, b] \rightarrow \mathbb{R}$

$$L = \int_a^b \sqrt{1 + f'(x)^2} dx$$

④. Obseg astroide

$$x^{2/3} + y^{2/3} = a^{2/3} \quad a > 0$$

$$x(t) = a \cos^3 t \quad t \in [0, 2\pi]$$

$$y(t) = a \sin^3 t$$

$$\dot{x}(t) = -3a \cos^2 t \sin t$$

$$\dot{y}(t) = 3a \cos t \sin^2 t$$

$$\begin{aligned} \sin t &= u \\ \cos t dt &= du \end{aligned}$$

$$\begin{aligned} L &= 4 \int_0^{\pi/2} 3a \sqrt{\cos^4 t \sin^2 t + \cos^2 t \sin^4 t} dt = 12a \int_0^{\pi/2} \cos t \sin t \sqrt{\cos^2 t + \sin^2 t} dt = \\ &= 12a \int_0^{\pi/2} \cos t \sin t dt = 12a \int_0^{\pi/2} u du = 6a (\sin^2 t) \Big|_0^{\pi/2} = 6a \end{aligned}$$

5. $f(x) = \sin x \quad x \in [0, \frac{\pi}{2}]$

dožina preko trapezne metode, napaka $< 0,01$

$$l = \int_0^{\frac{\pi}{2}} \sqrt{1 + \cos^2 x} dx =$$

$$= \frac{\pi}{8} \left(\frac{y_0 + y_8}{2} + y_1 + \dots + y_7 \right)$$

x	$\sqrt{1 + \cos^2 x}$
0	$\sqrt{2}$
$\frac{\pi}{16}$	1,401
$\frac{\pi}{8}$	1,361
$\frac{3\pi}{16}$	1,301
$\frac{\pi}{4}$	1,225
$\frac{5\pi}{16}$	1,144
$\frac{3\pi}{8}$	1,071
$\frac{7\pi}{16}$	1,019
$\frac{\pi}{2}$	1

$$= \frac{\pi}{16} \left(\frac{1 + \sqrt{2}}{2} + \dots \right) \approx 1,9101$$

Vrtenine:

$$V = \pi \int_a^b f^2(x) dx$$

$$P = 2\pi \int_a^b f(x) \sqrt{1 + f'(x)^2} dx$$

$$|R_n| \leq \frac{(b-a)^3}{12n^2} \max(g'(x))_{[0, \frac{\pi}{2}]} \leq 0,01$$

$$g(x) = \sqrt{1 + \cos^2 x}$$

$$g'(x) = \frac{1}{2} (1 + \cos^2 x)^{-\frac{1}{2}} (2 \cos x \sin x) (-1)$$

$$g''(x) = -\frac{1}{2} \left((-\frac{1}{2}) (1 + \cos^2 x)^{-\frac{3}{2}} (\sin 2x) \cdot 2 + \right.$$

$$\left. + (1 + \cos^2 x)^{-\frac{1}{2}} 2 \cos 2x \right) =$$

$$= \frac{1}{2} \frac{\sin 2x}{\sqrt{(1 + \cos^2 x)^3}} - \frac{\cos 2x}{\sqrt{1 + \cos^2 x}} \approx \frac{7}{4}$$

$$\frac{(\frac{\pi}{2})^3}{12n^2} \frac{7}{4} \leq \frac{1}{100}$$

$$\frac{7\pi^3}{48n^2 \cdot 4} \leq \frac{1}{100}$$

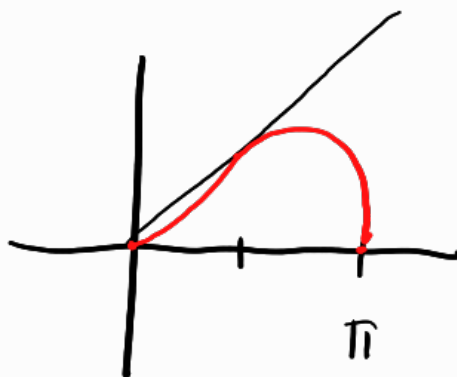
$$n^2 \geq \frac{100\pi^3}{48 \cdot 4}$$

$$n \geq 8$$

6. $V = ?$

$$f: [0, \pi] \rightarrow \mathbb{R}$$

$$f(x) = 2x \sin x$$



$$V = 4\pi \int_0^{\pi} x^2 \sin^2 x \, dx = 4\pi \int_0^{\pi} x^2 \frac{1 - \cos 2x}{2} \, dx =$$

$$= 2\pi \left[\frac{x^3}{3} \right]_0^{\pi} - 2\pi \int_0^{\pi} x^2 \cos 2x \, dx = \frac{2\pi^4}{3} - 2\pi \left(\left[\frac{x^2 \sin 2x}{2} \right]_0^{\pi} - \int_0^{\pi} x \sin 2x \, dx \right) =$$

$$\begin{aligned} \cos 2x \, dx &= du_1, & x^2 &= u_1 \\ \frac{\sin 2x}{2} &= v_1, & 2x \, dx &= du_1 \end{aligned}$$

$$\begin{aligned} \sin 2x \, dx &= du_2, & x &= u_2 \\ -\frac{\cos 2x}{2} &= v_2, & dx &= du_2 \end{aligned}$$

$$= \frac{2\pi^4}{3} + 2\pi \left(- \left[\frac{x \cos 2x}{2} \right]_0^{\pi} + \int_0^{\pi} \frac{\cos 2x}{2} \, dx \right) =$$

$$= \frac{2\pi^4}{3} + 2\pi \left(- \frac{\pi}{2} + 0 \right) = \frac{2\pi^4}{3} - \pi^2$$